

Study Report: Python Programming for Beginners - Learn Coding in 7 Days

Written by Gagan Pasupuleti

Family:	Multi-Language Programming
Level:	Beginner
Chapters:	8
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Summary

This report covers a seven-day plan to learn Python coding. It sets up the environment and the Python shell, starts programming, then covers variables and expressions, functions, loops, dictionaries/lists/tuples, object-oriented programming with inheritance, and error handling with try and except. It offers a compact, structured path for beginners to gain working skills in a week.

Chapters

Chapter 1: Setting Up Your Environment

Chapter focus: Setting Up Your Environment

Before writing Python programs, you install the Python interpreter from python.org (choose Python 3). During setup on Windows, check 'Add Python to PATH' so you can run python from the terminal. After installation, open a terminal and type `python --version` to confirm. You can write code in any text editor, but beginners often use IDLE (included with Python) or VS Code. The interactive shell (`>>>` prompt) is useful for testing one line at a time; saved `.py` files are for complete programs.

Code Reference:

```
import sys
print(sys.version)
print("Setup OK")
```

What it shows: import sys loads system info; sys.version shows your Python version.

Topic guide

What it actually is

Installation means putting the Python interpreter and tools on your computer so it can read and execute `.py` files. PATH is a list of folders Windows/Mac search when you type a command.

When to use it

Install Python once on any machine where you plan to write or run Python code. Reinstall or upgrade when you need a newer version for a library or course.

Where to use it

On your laptop for learning, on a server for web apps, on a Raspberry Pi for hardware projects, or in cloud notebooks like Google Colab (no local install needed there).

Real use example

A student installs Python 3.12, opens VS Code, creates hello.py, runs it from the terminal, and sees 'Setup OK' - confirming the environment works before starting homework.

Chapter 2: Let's Start Programming

Chapter focus: Let's Start Programming

Python is a general-purpose programming language known for readable syntax and wide use in web development, automation, data analysis, and machine learning. Unlike many languages that use braces and semicolons, Python uses indentation to show structure, which helps beginners see how code is organized. You write instructions in a .py file or run them line by line in the interactive shell. Each instruction tells the computer to do something: store a value, print text, make a decision, or repeat a task.

Code Reference:

```
print("Hello, Python!")  
print(2 + 3)
```

What it shows: The first line prints text. The second shows Python can also calculate numbers.

Topic guide

What it actually is

Python is a high-level, interpreted language: you write human-readable code and the Python interpreter runs it without a separate compile step. It comes with a large standard library and thousands of third-party packages.

When to use it

Use Python when you want to build something quickly, automate repetitive tasks, analyze data, create a web backend, or learn programming for the first time.

Where to use it

Used at Google, Netflix, NASA, and in schools worldwide. Common in data science teams, DevOps automation, scripting, and beginner coding courses.

Real use example

A small business owner writes a 10-line Python script that renames 200 invoice PDFs by date every month, saving an hour of manual work.

Chapter 3: Variables, Expressions, and Instructions

Chapter focus: Variables, Expressions, and Instructions

A variable is a name that points to a value stored in memory. You create one with the assignment operator (=): the name goes on the left, the value on the right. Python figures out the type automatically - you do not declare int or str first. Common types are integers (whole numbers), floats (decimals), strings (text in quotes), and booleans (True/False). You can change a variable later by assigning a new value. Clear names like `user_age` or `total_price` make code easier to read than single letters.

Code Reference:

```
count = 5          # integer
label = "box"      # string
price = 2.5        # float
is_open = True     # boolean
print(type(count)) # <class 'int'>
```

What it shows: Each variable stores one value. `type()` tells you what kind of data it holds.

Topic guide**What it actually is**

Variables are labeled containers for data. The label (name) lets you reuse and update values throughout a program without repeating the actual number or text.

When to use it

Use variables whenever a value might change, will be reused later, or needs a meaningful name - scores, user names, prices, flags, counters.

Where to use it

Every Python program: games (player score), web apps (username), data scripts (file path), calculators (operands), and loops (counters).

Real use example

An online shop stores `item_price = 499`, `quantity = 2`, and computes `total = item_price * quantity` so the checkout page shows Rs 998 without hard-coding numbers.

Chapter 4: Functions**Chapter focus: Functions**

Keep functions small and named after what they do. Document tricky ones with a one-line docstring. Return early when possible to avoid deep nesting.

Code Reference:

```
def area(width, height):
    return width * height

def print_area(w, h):
    print(area(w, h))
```

```
print_area(4, 5) # 20
```

*What it shows: area computes and returns; print_area reuses area instead of duplicating $w * h$.*

Topic guide

What it actually is

A function is a named, reusable mini-program inside your program. You call it by name with arguments; it may return a result.

When to use it

Use a function when the same logic appears twice, when a task has a clear name, or when you want to test one piece separately.

Where to use it

Calculations, formatting output, validating input, API handlers, and splitting large scripts into readable pieces.

Real use example

A password checker function `validate(pw)` returns True/False and is reused on signup, login, and reset forms.

Chapter 5: Loops

Chapter focus: Loops

Choose for when you know how many items or iterations; choose while when waiting for a condition (valid input, game running). Nested loops handle grids and tables.

Code Reference:

```
total = 0
for n in [10, 20, 30]:
    total += n
print(total) # 60

for i in range(3):
    print("Try", i + 1)
```

What it shows: First loop sums list values; second uses range for a fixed number of repetitions.

Topic guide

What it actually is

A loop is a control structure that repeats execution. `for` is for known iterations; `while` is for repeat-until-done patterns.

When to use it

Use loops to process every item in a collection, repeat until valid input, run a game main loop, or accumulate totals.

Where to use it

Reading files line by line, batch renaming files, training epochs in ML, printing tables, polling sensors on Arduino.

Real use example

A multiplication table uses nested for loops over rows and columns to print 1x1 through 10x10.

Chapter 6: Dictionaries, Lists, and Tuples

Chapter focus: Dictionaries, Lists, and Tuples

Tuples use parentheses, keep order, and cannot be changed after creation - ideal for coordinates, RGB colors, and database rows returned from queries. Lists are the default flexible container. Common methods: append, extend, insert, pop, remove, sort, reverse. Slicing `list[1:4]` copies a portion without affecting the whole list.

Code Reference:

```
scores = [88, 92, 75]
scores.append(90)
scores.sort()
print(scores[0], scores[-1]) # 75 92
```

What it shows: append adds an item; sort orders the list; [0] is first, [-1] is last.

Topic guide

What it actually is

A list is a mutable, ordered collection. Order matters: `['a','b']` is different from `['b','a']`.

When to use it

Use a list when you have many values of the same kind and need to add, remove, sort, or loop through them.

Where to use it

Shopping cart items, test scores, lines read from a file, todo tasks, or collecting results in a loop.

Real use example

A todo app keeps tasks in a list, appends new items, and removes completed ones with `pop()`.

Chapter 7: Object-Oriented Programming and Inheritance

Chapter focus: Object-Oriented Programming and Inheritance

Object-oriented programming models real things as objects with data (attributes) and behavior (methods). A class is the blueprint; `__init__` sets up new objects. `self` refers to the current instance. Inheritance lets a child class reuse and extend a parent class. OOP helps when many objects share the same structure - users, products, bank accounts.

Code Reference:

```
class BankAccount:
    def __init__(self, owner, balance=0):
```

```

        self.owner = owner
        self.balance = balance
    def deposit(self, amount):
        self.balance += amount

acc = BankAccount("Mia", 100)
acc.deposit(50)
print(acc.balance)  # 150

```

What it shows: `__init__` sets owner and balance; deposit updates balance on the object acc.

Topic guide

What it actually is

OOP organizes code around objects - bundles of data and functions that operate on that data.

When to use it

Use classes when you have many similar entities, need clear structure in large programs, or model real-world things (Student, Order, Sensor).

Where to use it

Games (Player, Enemy), apps (User, Post), GUIs, and libraries like Django models.

Real use example

A school app defines class Student with name and grades; each student object stores its own data while share methods like `add_grade()` work the same for everyone.

Chapter 8: Error Handling

Chapter focus: Error Handling

Log the error for developers but show a friendly message to users. finally: runs cleanup (close files) whether or not an error occurred.

Code Reference:

```

try:
    age = int(input("Age: "))
    print(100 / age)
except ValueError:
    print("Please enter a whole number.")
except ZeroDivisionError:
    print("Age cannot be zero.")

```

What it shows: Each except catches a specific error so the user gets a clear message.

Topic guide

What it actually is

An exception is Python's signal that something went wrong at runtime. Handling means catching that signal and recovering gracefully.

When to use it

Use try/except around user input, file access, network calls, and int/float conversions.

Where to use it

Forms, file importers, API clients, calculators, and any code that can receive bad data.

Real use example

A bank transfer uses try/except around the API call; on failure it logs details and shows 'Transfer failed, try again.'